

TOMÁS LOPERA DUQUE

Data Engineer | Data Scientist | AI Engineer | Software Engineering Student | AWS

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EXPERIENCE

Data & AI Engineer Intern

Cadena SA • Medellín, Colombia

Jan 2026 – Jul 2026

- Built an ETL pipeline using AWS Textract to extract structured data from physical forms and load it into a relational database, replacing a fully manual data entry process.
- Designed an AWS architecture using SageMaker to automate the training workflow of a YOLO object detection model, including data ingestion, processing job orchestration, and output pipelines.
- Migrated a legacy Excel macro-based ETL process to Python and deployed a cloud data warehouse architecture using Amazon Redshift, with self-service dashboards consumed by the analytics team via Amazon QuickSight.

Cloud Developer

Cuántico Seguros • Medellín, Colombia

Apr 2026 – Jun 2026

Developed and deployed a web platform for an insurance company, with ownership over both frontend development and AWS cloud infrastructure using S3, Cloudfront, Contentful, Google Analytics and Frontend Design.

AI Developer

Universidad EIA • Medellín, Colombia

Jun 2025 – oct 2025

Developed an OCR MVP and proposed a full AWS deployment architecture as part of a university-industry innovation challenge for EIA University and Cadena SA.

AWARDS & ACHIEVEMENTS

Hackathon Winner | AI Negotiation System

EIA University x Sistecrédito

Mar 2025

Performed data cleaning and EDA on customer financial data to build the data preparation layer supporting a negotiation assistant. **Stack:** Python, Pandas, OpenAI.

Hackathon Runner-Up | Logistics Agent

EIA University x Haceb

Aug 2025

Built a natural language logistics agent for Haceb's call center, enabling order and inventory queries via chatbot using RAG architecture and LLM response generation, orchestrated with n8n. **Stack:** n8n, Supabase, PostgreSQL, OpenAI

EDUCATION

Software & Computing Engineering

EIA University • Envigado, Colombia

Jan 2022 – Dec 2026

Focus: Data Engineering & Data Science / AI

SUMMARY

Data Engineer with hands-on experience building ETL pipelines, cloud data warehouses, and automated data workflows on AWS. Delivered production solutions at Cadena SA using SageMaker, Glue, Redshift, and Textract. Complementary experience in AI development and cloud infrastructure deployment. Strong foundation in pipeline orchestration, data modeling, and scalable cloud architecture. Graduating December 2026 from EIA University.

SKILLS

Data Engineering

- SQL & Data Manipulation
- Python for data processing and automation
- ETL/ELT Pipeline Development & Orchestration
- Data Warehouse, Lake & Lakehouse Modeling
- Pipeline Orchestration & Workflow Automation

Data Science & AI

- Exploratory Data Analysis & Visualization
- Feature engineering and preprocessing
- Model Training & Evaluation
- Generative AI & Large Language Models (LLMs)

Tech Stack

- Languages & Core. Python • SQL • Git
- AWS. IAM • EC2 • S3 • RDS • Lambda • Glue • Athena
- Redshift • SageMaker • Textract
- Data Engineering. Pandas • Spark • dbt • Prefect • Databricks • Snowflake • Airflow
- Data Science & AI. scikit-learn • Langchain
- Matplotlib • Seaborn
- Other. n8n • FastAPI • Docker • PowerBI

LANGUAGES

Spanish (Native) ●●●●●
English (B2+) ●●●●○

CERTIFICATIONS

Deep Learning Fundamentals

NVIDIA, 2025

Transformed-Based NLP Apps

NVIDIA, 2025

LLM Apps with Prompt Engineering

NVIDIA, 2025